

Interpretability Analysis using Partial Dependence Plots (PDP)

Authors: Chenyao Lin, Antonio Sánchez Guillén

1. Introduction

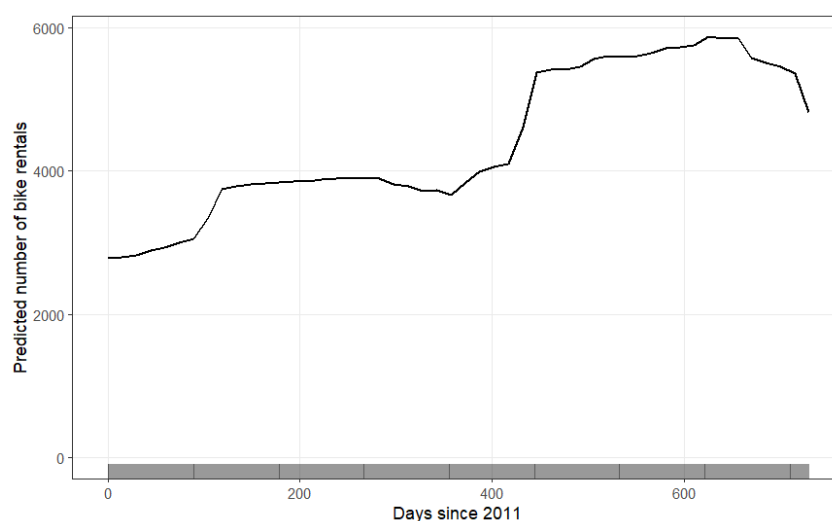
Partial Dependence Plots (PDP) are a model-agnostic interpretability tool that shows the marginal effect of one or two features on the predicted outcome of a trained model, averaging out the contribution of all other variables. This report applies PDPs to two datasets using Random Forest regressors: the UCI Bike Sharing dataset (day.csv) and the King County house price dataset (kc_house_data.csv).

2. Bike rentals

The dataset contains 731 daily observations. Features were preprocessed before training: temperature was denormalized to °C ($\text{temp} = \text{temp} * (39 - (-8)) + (-8)$), humidity to percentage ($\text{hum} = \text{hum} * 100$), and wind speed to km/h ($\text{windspeed} = \text{windspeed} * 67$). A new variable `days_since_2011` was computed as the number of days elapsed from 1 January 2011. The model uses only these four numeric features plus `cnt` as the target.

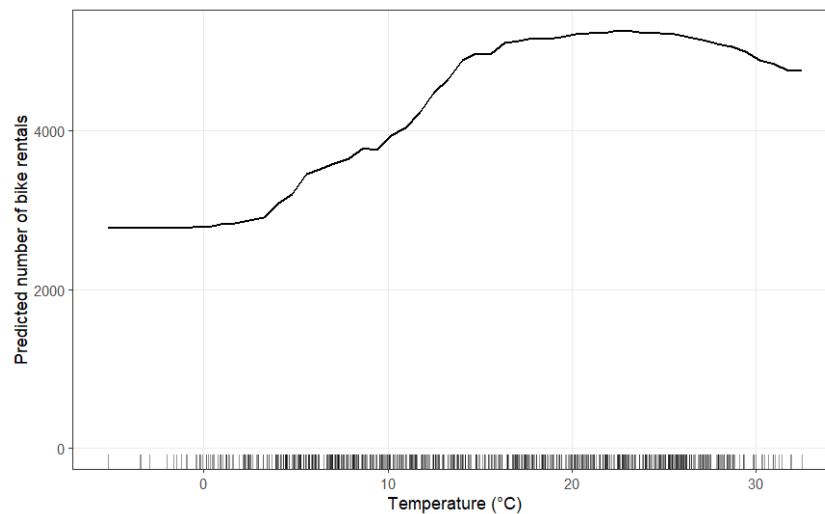
A Random Forest with 500 trees was trained on the full dataset. PDPs were computed by marginal averaging over a 50-point grid for each feature, substituting each grid value across all rows and averaging the predicted output.

2.1. 1D Partial Dependence Analysis

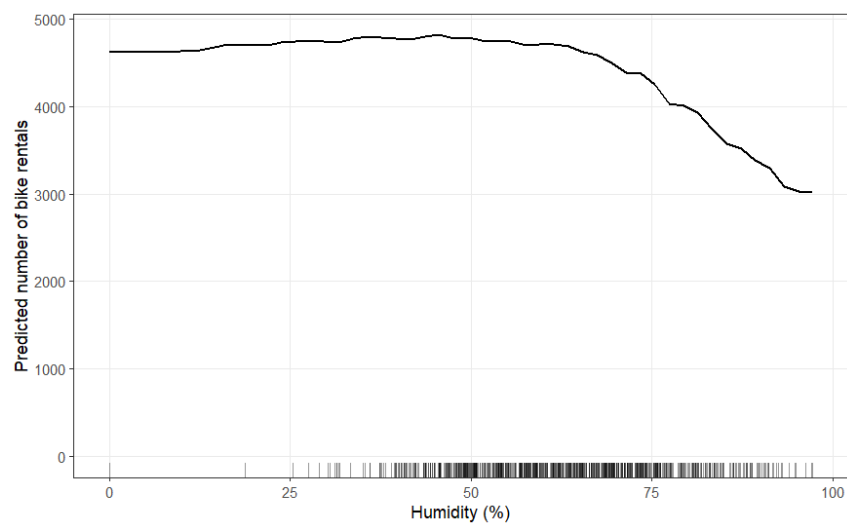


Days since 2011 shows the strongest effect. Predicted rentals grow from roughly 2700 at the start of 2011 to about 5800 by mid-2012, reflecting the year-on-year growth in service adoption. The curve is not perfectly smooth: a step around day 100

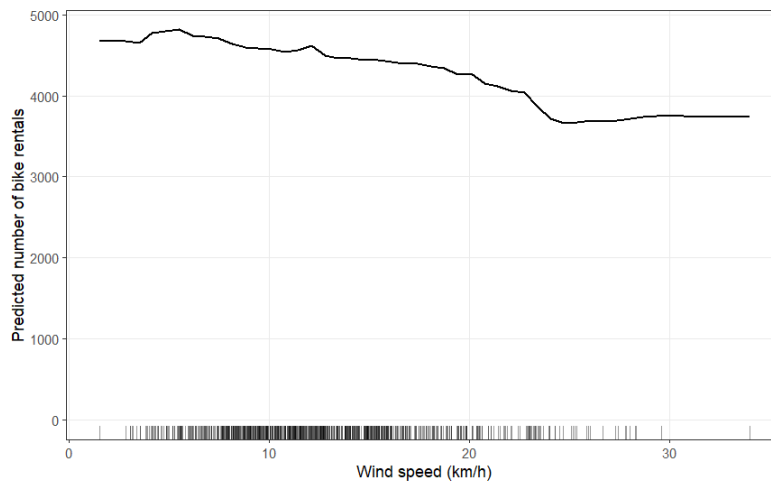
corresponds to Spring 2011, and the steeper climb after day 400 marks the start of the second year.



Temperature has a clear positive effect up to roughly 15°C, after which predictions stabilise around 5,500 and decline slightly above 25°C. Cold temperatures suppress demand to about 2,800 per day. The shape is consistent with comfortable cycling conditions existing in a moderate temperature range.



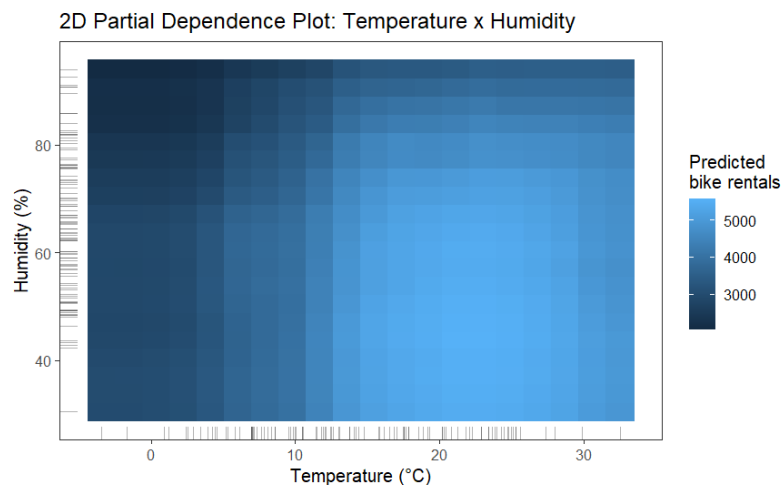
Humidity has little effect below 60-70%, with predicted rentals stable around 4,600. Above 70%, the curve drops to 3,000 at near-maximum humidity. High humidity is associated with rainy or oppressive conditions that discourage cycling.



Wind speed has a gradual negative effect on predicted bike rentals. Demand starts near 4,700 rentals at low wind speeds and declines steadily to around 3,700 as wind exceeds 25 km/h, after which the curve stabilises. Strong winds make cycling physically demanding and very high values are rare in the data, which contributes to the levelling-off at the tail.

2.2. 2D Interaction Analysis

To limit computation time, a random sample of 100 observations was drawn from the dataset. A 20×20 grid was built over the observed ranges of temperature and humidity. For each grid cell, both features were simultaneously substituted across all sampled rows and the mean prediction was recorded. Tile dimensions were slightly inflated to avoid gaps in the rendered plot.

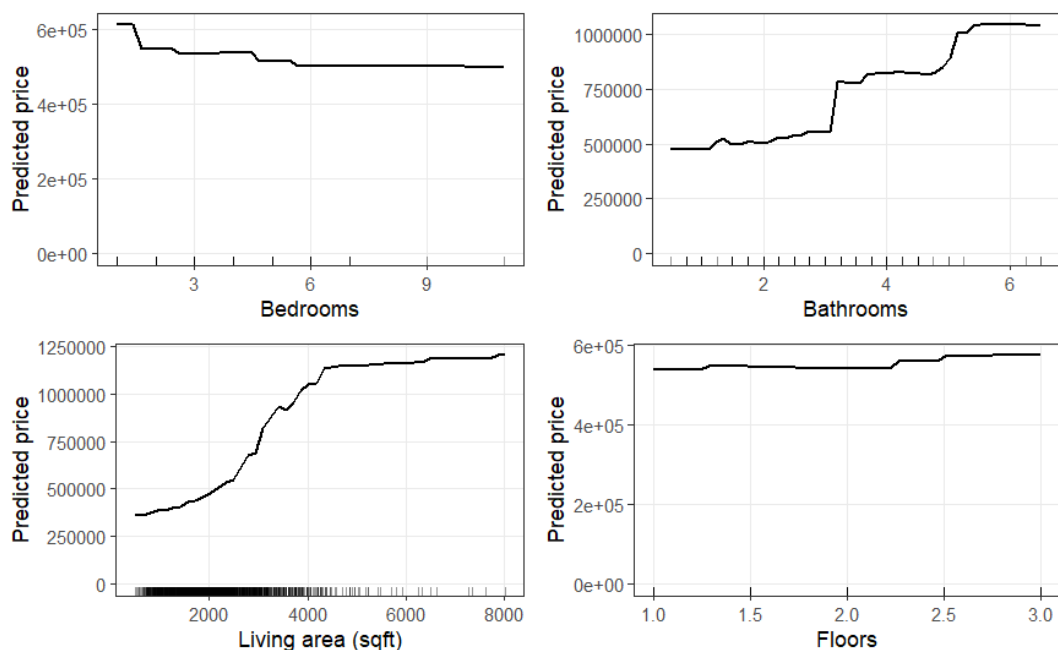


Temperature is clearly the dominant driver of bike rentals in this interaction. Predicted demand increases substantially as temperatures rise, regardless of humidity level. Humidity, by contrast, exerts a modest suppressing effect: higher humidity is associated with slightly lower predicted rentals, but this influence is secondary and relatively uniform across temperature values. The highest predicted demand is concentrated in the warm, lower-humidity region of the plot (bottom-right), while the lowest demand corresponds to cold and high-humidity conditions (top-left).

3. House Prices

Due to the dataset's volume (over 21,000 records), and following computational efficiency best practices for interpretability methods, a random sample of 1,500 instances was extracted. A Random Forest Regressor model was trained using the following variables: bedrooms, bathrooms, sqft_living, sqft_lot, floors, and yr_built. Subsequently, 1D and 2D PDPs were generated to analyze the influence of the four requested main variables.

3.1. 1D Partial Dependence Analysis



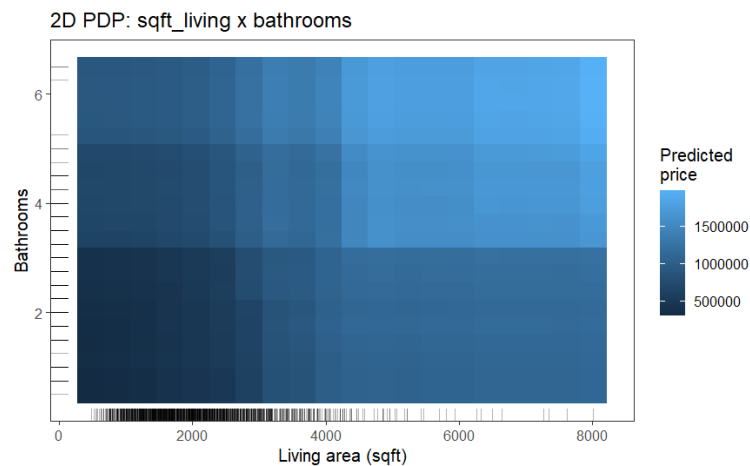
Living area (sqft) is the strongest individual predictor of price. The relationship is strongly positive and roughly monotonic: predicted prices rise from approximately \$350,000 for the smallest homes to around \$1,200,000 for the biggest properties. The steepest gains occur between 2,000 and 4,000 sqft, after which the curve begins to flat, suggesting diminishing marginal returns at the high end of the distribution.

The number of bathrooms shows a clear positive association with predicted price, with values rising from roughly \$500,000 at 1 or 2 bathrooms to over \$1,000,000 at 6+ bathrooms. The relationship follows a step-wise pattern rather than a smooth curve. The most notable jumps occur between 3 and 4 bathrooms.

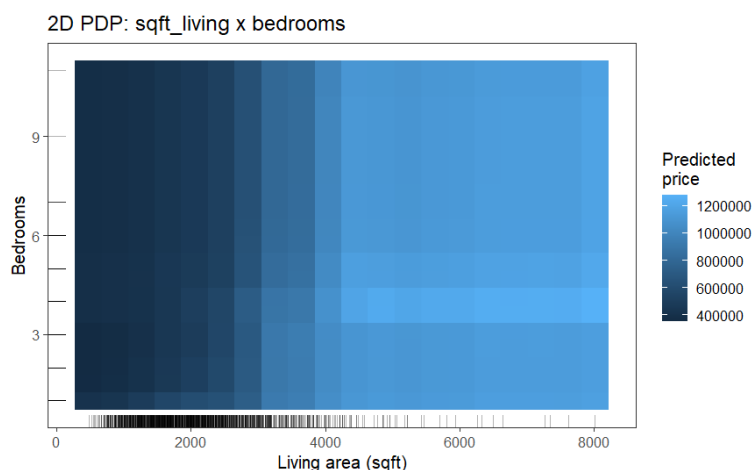
Contrary to intuitive expectations, bedrooms exhibit a negative partial dependence: predicted prices decline as bedroom count increases, dropping from over \$600,000 at 1 bedroom to around \$500,000 at 6+ bedrooms, where the curve stabilises.

The number of floors have a comparatively flat effect on predicted price. Values remain fairly stable between approximately \$500,000 and \$600,000 across the full range (1 to 3 floors), with only a modest upward trend. This suggests that the number of floors contributes little additional predictive power beyond what is already captured by living area.

3.2. 2D Interaction Analysis



The interaction plot confirms that both living area and bathroom count exert independently positive effects on predicted price and that these effects are largely additive. The highest predicted prices (>\$1,500,000) appear in the upper-right corner, corresponding to large homes with many bathrooms. Notably, the gradient runs more strongly along the horizontal axis (living area) than the vertical axis (bathrooms), reinforcing that living area is the dominant driver.



This plot reveals a nuanced interaction between living area and bedroom count. The dominant pattern is again driven by living area, but the effect of bedrooms varies depending on the size of the home. For smaller properties (below 3,000 sqft), higher bedroom counts are associated with lower predicted prices, likely because more bedrooms imply smaller individual rooms given a fixed square footage. For larger

properties, this penalty is less pronounced, suggesting that in spacious homes, additional bedrooms are less detrimental, possibly because they do not imply cramped room sizes in that context. Notably, the highest predicted prices are concentrated in homes exceeding 4,000 sqft with around 4 bedrooms, a combination that appears to strike an optimal balance between spaciousness and bedroom count.

4. Conclusions

Partial Dependence Plots have been applied to two distinct datasets to examine how individual features and their interactions shape model predictions.

For bike rentals, the analysis reveals that temporal growth (days since 2011) is the single strongest driver of demand, reflecting the progressive adoption of the service over time. Temperature exerts a clear positive effect up to moderate values, after which demand plateaus and slightly declines, consistent with the physical comfort constraints of cycling. Humidity shows little influence below 70%, but suppresses demand markedly at higher levels, while wind speed produces a gradual and steady negative effect. The 2D interaction between temperature and humidity confirms that these two variables act largely independently, with temperature dominating and humidity playing a secondary, uniformly suppressing role.

For house prices, living area emerges as the dominant predictor, with a strong positive and roughly monotonic relationship that shows diminishing returns beyond 4,000 sqft. Bathrooms contribute positively in a step-wise fashion, while floors have a comparatively negligible effect. Bedrooms, contrary to intuition, exhibit a negative partial dependence, a result best explained by the bedroom-to-space ratio: in smaller properties, more bedrooms imply cramped rooms, which the model penalises. The 2D interaction analyses confirm that living area consistently drives the gradient more strongly than any other variable examined, and that the most valuable properties tend to be those combining large floor area with a moderate to high bathroom count.

Overall, PDPs prove to be an effective and interpretable tool for understanding feature contributions in Random Forest models, provided their assumption of feature independence is kept in mind when drawing conclusions from interaction plots.

References

Ferri, C., & Monserrat, C. (2026). *T3: Reliable models - Evaluation, deployment and monitoring of models*. Degree in Data Science, ETSInf, Universitat Politècnica de València.

Fanaee-T, H., & Gama, J. (2013). *Bike sharing dataset* [Data set]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5W894>

KC House Data Dataset. King County Residential Home Sales Database.